ASSIGNMENT–1

Foundations of Cryptography

Course: BCA | Subject: Cryptography

Q1. Define cryptography in your own words. What are the three main goals it tries to achieve?

Cryptography is the technique of protecting information by transforming it into a form that unauthorized people cannot understand or misuse. Its three main goals are confidentiality (keeping information secret), integrity (ensuring information is not altered), and authentication (verifying the identity of the communicating party).

Q2. Differentiate between plaintext and ciphertext with a simple example.

Plaintext is the original readable information before encryption, while ciphertext is the scrambled form produced after encryption. For example, if the plaintext is “HELLO” and a simple Caesar shift of 3 is used, the ciphertext becomes “KHOOR”.

Q3. Explain the concept of cryptography in detail. Discuss why it is considered essential in the modern digital world, giving at least three real-life applications.

Cryptography is a security technique that uses algorithms and keys to protect information during storage and communication. Encryption converts readable plaintext into ciphertext, while authorized users can use the appropriate key to recover the original information. Modern cryptography also supports integrity checking, authentication, and digital signatures.

Cryptography is essential because digital information frequently travels through networks where unauthorized parties may attempt to read, modify, or impersonate users. It protects sensitive data and helps establish trust between communicating systems.

Real-life applications include:
1. Banking: online banking and payment systems use encryption to protect account details and financial transactions.
2. Messaging apps: end-to-end encryption helps prevent unauthorized parties from reading private messages.
3. E-commerce: cryptographic protocols protect customer information and payment-related communication between browsers and online stores.

Q4. What is the difference between encryption and decryption?

Encryption converts plaintext into unreadable ciphertext using an algorithm and key so that unauthorized people cannot understand the information. Decryption is the reverse process, in which ciphertext is converted back into plaintext using the appropriate key.

Q5. Describe Symmetric Key Cryptography and Asymmetric Key Cryptography. Compare them based on key usage, speed, security, and typical use cases.

Symmetric key cryptography uses the same secret key for both encryption and decryption. The communicating parties must securely share this key before protected communication can take place. It is generally fast and is therefore suitable for encrypting large amounts of data. Examples include AES and DES.

Asymmetric key cryptography uses a pair of mathematically related keys: a public key and a private key. The public key can be shared openly, while the private key is kept secret. It is generally slower than symmetric cryptography but solves important key-distribution and authentication problems. Examples include RSA and ECC.

Comparison:
• Key usage: Symmetric uses one shared secret key; asymmetric uses a public/private key pair.
• Speed: Symmetric algorithms are generally faster; asymmetric algorithms are generally slower.
• Security: Both can provide strong security when correctly implemented, but they address different problems. Asymmetric cryptography is especially useful for secure key exchange and digital signatures.
• Typical use: Symmetric cryptography is used for bulk data encryption, while asymmetric cryptography is commonly used for key exchange, authentication, and digital signatures.

Modern systems often combine both approaches: asymmetric cryptography establishes or protects a session key, and symmetric cryptography efficiently encrypts the actual data.

Q6. What is hashing, and how is it different from encryption?

Hashing converts data into a fixed-length hash value using a hash function and is normally designed to be one-way. Encryption is designed to protect data in a reversible way so that authorized users can recover the original information using a key; hashing is commonly used for integrity checking and secure password storage rather than recovering the original data.

Q7. Critically evaluate why achieving confidentiality, integrity and non-repudiation together is challenging in a real-world communication system.

Confidentiality, integrity, and non-repudiation require different security mechanisms and careful key management. Encryption can provide confidentiality, integrity can be supported by authentication mechanisms such as MACs or authenticated encryption, and non-repudiation generally requires digital signatures and trustworthy key ownership. Combining these properties also creates practical challenges involving certificate management, private-key protection, performance, interoperability, and secure endpoint implementation.

Q8. Explain the role of cryptography in network security. Discuss how it helps achieve confidentiality, integrity, authentication, and non-repudiation, with one example for each.

Cryptography is a fundamental part of network security because information moving across networks may be exposed to interception, modification, or impersonation.

• Confidentiality: Encryption prevents unauthorized parties from understanding transmitted data. Example: HTTPS/TLS encrypts web traffic between a browser and a server.
• Integrity: Cryptographic integrity mechanisms detect unauthorized modification of data. Example: a message authentication mechanism can detect whether a transmitted message has been changed.
• Authentication: Certificates, public-key cryptography, and cryptographic protocols help verify the identity of communicating parties. Example: a TLS certificate helps a browser authenticate the intended website.
• Non-repudiation: Digital signatures provide evidence that a message or document was signed using a particular private key. Example: a digitally signed document can be verified using the corresponding public key.

Together, these mechanisms help create trusted and secure network communication.

Q9. Name any two network security protocols that use cryptographic techniques and briefly state their purpose.

1. SSL/TLS: Protects application-layer communication, especially web traffic, by providing encryption, integrity protection, and authentication.
2. IPsec: Secures IP communications by providing cryptographic protection for network-layer traffic, including confidentiality and integrity mechanisms.

Q10. Critically discuss the evolution of cryptography from classical ciphers to modern cryptographic systems, evaluating the impact of computational advances on cryptographic security.

Classical cryptography relied mainly on substitution and transposition techniques, such as the Caesar and Vigenère ciphers. These methods were useful historically but their relatively small key spaces and structural weaknesses made them vulnerable to systematic analysis.

With computers, cryptography became more mathematical and algorithmically complex. Systems such as DES introduced computer-oriented symmetric encryption, while RSA introduced practical public-key cryptography. AES later replaced DES as a modern standard for efficient symmetric encryption.

Computational advances have had two opposite effects. They allow stronger algorithms to be designed and deployed efficiently, but they also give attackers greater processing power for brute-force searches, cryptanalysis, and automated attacks. As a result, cryptographic systems must use sufficiently large keys, strong algorithms, secure implementations, and regular updates. Security therefore depends not only on the algorithm but also on key management, implementation, and the changing capabilities of attackers.

Q11. Trace the historical development of cryptographic techniques, starting from classical methods through modern computer-based cryptography. Highlight key turning points such as World War II cryptography and the invention of public-key cryptography.

Cryptography has developed from simple manual ciphers to complex computer-based security systems.

1. Classical period: The Caesar Cipher used a fixed shift of letters, while the Vigenère Cipher used a keyword to create a changing substitution pattern. These methods provided basic secrecy but were vulnerable to cryptanalysis.
2. World War II: The Enigma machine demonstrated the importance of complex mechanical and mathematical cryptography. Efforts to break Enigma contributed significantly to the development of modern cryptanalysis and computing.
3. Computer-based symmetric cryptography: DES became an important standardized symmetric encryption algorithm. It demonstrated the use of computers for practical secure data encryption, although its key size eventually became inadequate against increasing computational power.
4. Public-key revolution: Public-key cryptography introduced separate public and private keys, addressing major key-distribution problems and enabling practical digital signatures. RSA became one of the best-known public-key algorithms.
5. Modern era: AES became a widely used symmetric standard with stronger security and efficient performance. Modern systems commonly combine symmetric and asymmetric techniques, along with hashing, certificates, authenticated encryption, and digital signatures.

The overall development shows a shift from simple substitution and transposition toward mathematically based systems designed to withstand powerful computational attacks.

Q12. Who introduced the concept of public-key cryptography, and in which year?

The concept of public-key cryptography was introduced by Whitfield Diffie and Martin Hellman in 1976. Their work, known as Diffie–Hellman key exchange, established the foundation of modern public-key cryptography.